SPE-227296-MS

Double-Shell Microcapsules Containing Oxidizer for Controlled Gel Breaking

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

The development of microencapsulated persulfates represents a significant advancement in hydraulic fracturing operations, particularly for high-temperature reservoirs where controlled gel-breaking is essential. This study demonstrates an innovative encapsulation technique utilizing Acrylonitrile-Butadiene Styrene (ABS) polymer through polydimethylsiloxane (PDMS) induced coacervation and microfluidics approach to effectively encapsulate ammonium persulfate (APS) breakers.

Our comprehensive investigation examined multiple characteristics of the fabricated microcapsules, including their chemical design, active agent concentration, physical form, architectural design, controlled-release behavior, and fracturing fluid degradation efficiency. The findings reveal successful formation of micro-scale encapsulated particles featuring permeable exteriors and distinct core-shell configurations. The release kinetics were notably dependent on the proportion between the APS core and ABS shell components during synthesis.

This breakthrough technology provides precise control over gel-breaking timing while preventing premature viscosity reduction - a common challenge in high-temperature fracturing operations.

The ABS coacervation method proves to be a practical solution for creating temperature-resistant encapsulated breakers with adjustable release profiles. This approach significantly enhances operational efficiency by maintaining optimal fluid viscosity during critical phases of fracturing while ensuring complete gel degradation when required. The technology offers substantial improvements over conventional breaker systems, particularly in challenging high-temperature reservoir conditions where precise control over fluid properties is paramount for successful stimulation operations.

Introduction

Hydraulic fracturing has revolutionized global energy production by enabling economic oil recovery from previously inaccessible reservoirs, particularly unconventional formations such as shale and tight sandstone deposits [1], [2], [3], [4]. This well stimulation technique relies on injecting high-pressure fracturing fluids into subsurface formations to create extensive fracture networks that greatly enhance hydrocarbon flow pathways. The success of this technology depends critically on the performance of fracturing fluids, which must simultaneously fulfill multiple complex functions: effectively transmitting hydraulic energy to fracture the rock matrix, efficiently transporting proppant materials to maintain fracture conductivity, and subsequently degrade the fracturing gel in a controlled manner to permit optimal hydrocarbon flowback [5]. These demanding requirements have driven continuous innovation in fracturing fluid chemistry and engineering, with particular attention to the gel-breaking process that represents both a key operational step and a significant technical challenge in fracturing operations [6], [7].

The composition of modern fracturing fluids represents a sophisticated balance of water, viscosifying agents (typically guar gum or its derivatives), crosslinking compounds, and specialized chemical breakers [8], [9]. Among these components, the gel breaker system plays a particularly crucial role in determining operational success. These chemical agents are responsible for the controlled and complete degradation of the polymer network after the fracturing operation, enabling the fluid to flow back from the formation while leaving the proppant in place [10], [11]. The timing of this breaking process is absolutely critical - premature degradation disturbs proppant placement and fracture geometry, while delayed breaking can lead to significant formation damage and impaired hydrocarbon recovery [12]. This balance becomes especially challenging in high-temperature reservoir environments where conventional breaker systems often fail to provide the necessary control over degradation kinetics.

Persulfate compounds, particularly ammonium persulfate (APS) and potassium persulfate (KPS), have become industry-standard oxidative breakers due to their ability to generate highly reactive free radicals through thermal decomposition [13]. These radicals effectively cleave the polysaccharide backbone of guar-based polymers, reducing fluid viscosity and facilitating flowback. However, the key properties that make persulfates effective breakers are high-water solubility and temperature-dependent decomposition kinetics that also create significant operational challenges [12]. Incorporating oxidant gel breakers directly into the fracturing fluid system presents difficulties, especially under high-temperature conditions. This approach may cause the fluid's viscosity to decrease too quickly, leading to early sand settling in the wellbore and ultimately compromising the success of the fracturing operation [14]. This limitation has driven extensive research into controlled-release technologies that can modulate persulfate activity to match specific reservoir conditions and operational timelines.

Microencapsulation has emerged as a particularly promising approach for controlling breaker release kinetics [15], [16], [17]. This technology involves coating active breaker compounds with protective polymeric shells that regulate exposure to the fracturing fluid [18], [19]. The concept builds on well-established encapsulation techniques from pharmaceutical and specialty chemical industries, but presents unique challenges when applied to highly water-soluble compounds like persulfates in extreme downhole conditions [20]. Current encapsulation methodologies for fracturing applications primarily employ either fluidized bed coating or in-situ polymerization techniques [21], [22], [23]. Fluidized bed processes, while capable of producing consistent microcapsules, require substantial capital investment in specialized equipment and present significant scale-up challenges for field applications [24]. In-situ polymerization methods offer more flexible encapsulation options but often involve complex reaction schemes that may be difficult to control in industrial settings [23], [24].

The limitations of existing encapsulation approaches have stimulated interest in alternative techniques that can provide better control over release profiles while maintaining cost-effectiveness for large-scale oilfield applications. Coacervation, a phase separation-based encapsulation method, has attracted particular attention due to its relatively simple implementation and potential for reagent recovery [25]. This technique exploits controlled phase separation of polymers in solution to form discrete coacervate droplets that deposit on core materials [26]. For fracturing applications, acrylonitrile-butadiene-styrene (ABS) has shown promise as a shell material when combined with polydimethylsiloxane (PDMS) as a coacervating agent [25]. The ABS/PDMS system offers several advantages including tunable mechanical properties, good thermal stability, and compatibility with common oilfield chemicals [27]. Recent work has demonstrated successful encapsulation of persulfates using this approach, achieving delayed viscosity reduction at elevated temperatures while maintaining fracturing fluid viscosity above 50 mPa·s for 100 minutes at 70°C [25,27].

The practical implementation of microencapsulated breakers faces several technical hurdles that must be addressed. First, the potential for secondary formation damage from residual shell materials requires careful consideration [28], [29]. Even minor accumulations of polymeric residues in fracture networks or pore spaces can significantly impair conductivity and hydrocarbon recovery [30], [31], [32], [33], [34]. Second, encapsulated breakers must maintain their integrity under the extreme mechanical stresses encountered during fracturing operations, including high shear rates during pumping, abrasion during passage through wellbore, and compressive loads in the fracture network [35], [36], [37]. These challenges are accompanied by the economic constraints of field operations, where cost considerations often dictate technology adoption regardless of technical merits [38].

Pressure-responsive microcapsules represent an innovative approach to addressing some of these challenges [39]. Unlike conventional encapsulation systems that rely on diffusion or thermal degradation for release, these designs incorporate mechanical triggers that activate breaker release in response to specific downhole conditions [35]. The concept takes advantage of fracture closure stresses to rupture microcapsules at the optimal time for fluid breaking [36], [37]. This approach offers potential advantages in kinetics control and may reduce concerns about premature release. However, most reported studies have focused on relatively low-pressure conditions that don't fully represent downhole environments, highlighting the need for more rigorous evaluation under realistic stress conditions [40].

The consequences of inadequate breaking systems extend beyond immediate operational concerns to long-term production impacts. Field experience indicates that 30-90% of injected fracturing fluid typically remains in the formation, where it can cause various forms of damage including polymer invasion into matrix pores, proppant pack impairment, and relative permeability [30]. These damage mechanisms can reduce effective fracture conductivity by 30-80% according to various studies, with corresponding impacts on production rates and ultimate recovery [32], [33], [34]. While oxidative breakers have been the traditional solution for frack fluid degradation, their limitations (including uncontrolled reaction kinetics, temperature sensitivity, and incompatibility with other additives) have become increasingly apparent as operations move into more challenging reservoirs [41], [42], [43], [44].

This context underscores the importance of developing improved encapsulation systems for persulfate breakers. The current study focuses on ABS/PDMS coacervation as a promising alternative to existing encapsulation methods, with particular attention to performance under conditions representative of actual field operations. The research encompasses multiple aspects of microcapsule development including formulation optimization, physicochemical characterization, release kinetics analysis, and evaluation of breaking performance under simulated reservoir conditions. A key innovation involves systematic investigation of the coacervation process parameters to establish relationships between manufacturing conditions and microcapsule properties. The study also examines the mechanical stability of microcapsules under simulated downhole stresses and evaluates potential formation damage effects.

The broader significance of this research lies in its potential to improve the efficiency and environmental footprint of hydraulic fracturing operations. More reliable breaker systems can reduce the volume of fracturing fluid required for effective stimulation, minimize formation damage, and improve hydrocarbon recovery factors. These improvements have direct economic benefits through enhanced well productivity and reduced remediation costs, while also addressing environmental concerns associated with fluid retention in formations. The encapsulation approach developed in this study may also find applications in other areas of oilfield chemistry where controlled release of active components is desired, such as corrosion inhibitors, scale preventers, or enhanced oil recovery chemicals.

As the oil and gas industry continues to develop more challenging reservoirs and faces increasing scrutiny regarding operational efficiency and environmental impact, the importance of advanced fluid technologies will only grow. Microencapsulated breaker systems represent one important piece of this technological evolution, offering the potential for more precise control over fluid behavior in the complex downhole environment. The current work contributes to this ongoing development by providing a systematic evaluation of an alternative encapsulation approach that balances performance requirements with practical implementation considerations. Future research directions suggested by this work include investigation of hybrid encapsulation systems combining multiple release mechanisms, development of shell materials with improved environmental profiles, and field trials to validate laboratory findings under actual operating conditions.

In this study, we prepared an encapsulated breaker with a double-shell structure via combination of coacervation and encapsulation by microfluidics approach, using vinyl acrylate TMPTA as the outer impermeable shell of capsule and ABS as the internal permeable shell of APS core. The microstructure and encapsulation rate of the encapsulated breaker were analyzed. Compared with the existing encapsulated breaker technology, the novelty of this study is to improve the temperature resistance and induce the delayed release of encapsulated breakers. The ABS/TMPTA encapsulated breaker has the advantages of low cost, high efficiency, and sustainability. In the future, the release kinetics of encapsulated breakers should be improved by optimizing the preparation method.

Materials and methods

Materials

The Acrylonitrile butadiene styrene (ABS) was selected as the permeable shell material. The molecular weight of the above ABS ranged from 10 000 to 12 000 Da. Ammonium persulfate (APS), the main material of this study. The photocrosslinkable prepolymer resin of trimethylolpropane triacrylate (TMPTA) was selected as the impermeable shell material. The bis(2,4,6-trimethylbenzoyl)phenylphosphine oxide (BAPO) was selected as photoinitiator. ABS, APS, TMPTA, BAPO, anhydrous methylene dichloride (DCM), n-heptane, decane, polydimethylsiloxane (PDMS) (commercial grade, 200 cSt) was obtained from commercial sources. The reagents were used as received without further purification.

Preparation of microcapsules

Creation of the permeable single-shell microcapsules (SMCs). The procedure is described elsewhere [45]. The encapsulation of APS was carried out at room temperature of ~25 °C. At the first step, APS was grinned in a mil and sieved to the maximal particle size 80 µkm. Secondly, to prepare the shell solution, ABS (1 g) was dissolved in DCM (30 mL) till complete solvation while stirring (300-400 rpm) at RT for about 10 min. Then, grinded APS powder (0.8 g) was added in one portion to the shell solution in a 100-mL beaker under 350 rpm stirring. The stirring of the mixture was continuing for 10 min, followed by dropwise addition of 30 mL of PDMS to coacervate the ABS on the APS surface. Next, the coacervated mixture was rapidly poured into 250 mL of n-heptane or decane under stirring (450 rpm) and kept for 30 min to form the porous ABS shell over the APS core. The obtained precipitate of SMCs was filtered through the paper filter, washed with 100 mL n-heptane or decane for three times and dried in a drying oven at 50 °C for 3 h. The scheme of the coacervation process is represented on Figure 1.

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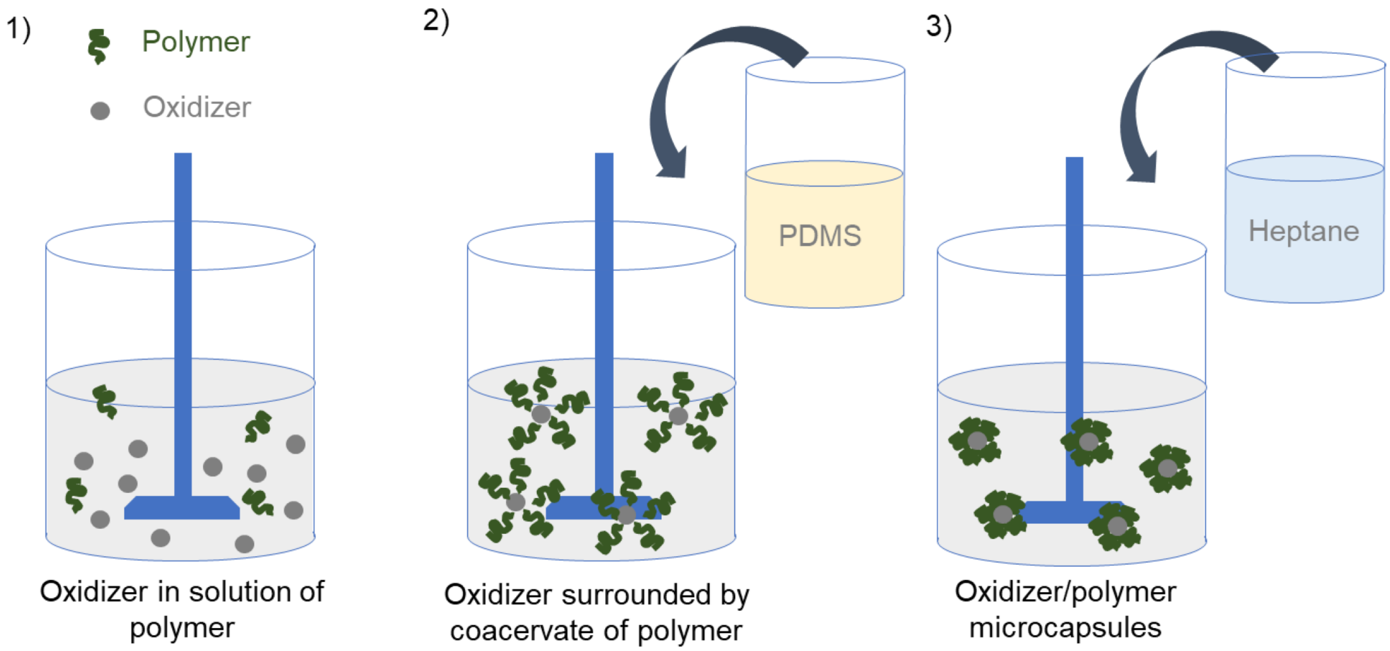


Figure 1—Scheme of the coacervation process for formation of microcapsules with core-shell structure.

Creation of the impermeable double-shell microcapsules (DMCs). The synthesis of DMCs is based on encapsulation of SMCs into polymer matrix with impermeable shell. It can be done by microfluidic approach. Triacrylate based photomonomer was used as matrix to be crosslink by photoinitiator BAPO resulting in impermeable outer shell. The procedure represented in Figure 2. The two liquid phases should be prepared. The first solution is inner phase; it consists of SMCs (2 g) dispersed in mixture of TMPTA (5 ml) and BAPO (5 wt.% of TMPTA). The Rhodamin B was added to TMPTA phase. The second solution is outer phase, which is a stabilized water (distilled water with 2-4 wt.% of polyvinyl alcohol).

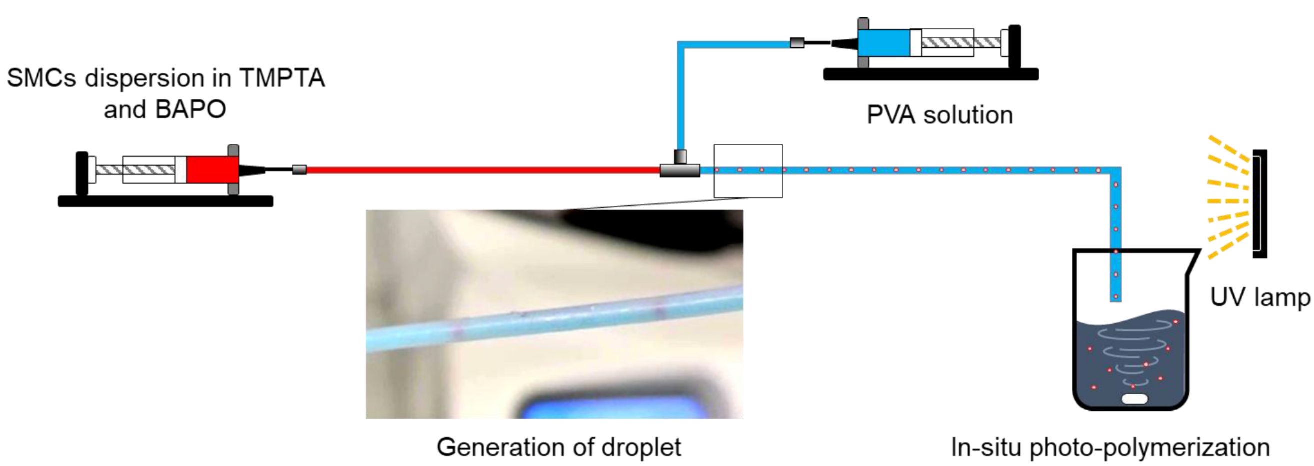


Figure 2—Microfluidic configuration for double-shell formation by UV-polymerization.

The both liquid phases were loaded into the syringes connected to each other by microfluidics tubes and one T-junction (Fig. 2. The inner and outer phase are depicted as red and blue syringe, respectively). The flows of both solutions through the microfluidics tubes were incorporated by two pumps. As a result, the droplets of inner phase in outer phase are formed. The flow of SMC dispersion in polymeric matrix being separated by outer phase passed through the UV-light irradiation zone to initiate the polymerization. The droplet's shell solidified by UV-light (wavelength of 400 nm) to form impermeable shell over the SMCs. The outlet of the microfluidics tubes accumulates DMCs in beaker. The DMCs collected by filtration through a paper filter and dried in air overnight.

Characterization

Chemical composition of microcapsules. The shells formation process is monitored by ATR-FTIR Spectroscopy.

Structure of microcapsules. The morphology of SMCs and DMCs was observed by optical microscopy. The SEM images of the SMCs, and DMCs were made on a field emission scanning electron microscope (Schottky cathode). The images obtained by SEM were used for the size determination and the distribution of constituents of SMCs and DMCs, thus elucidating the structure of SMCs and DMCs.

Capillary electrophoresis. Capillary electrophoresis is a method used to determine the content of cations (NH^{3+}, K^+, Na^+, Mg^{2+}, Ba^{2+}, Ca^{2+}) in water solution. It is implemented by separating the charged components of a complex mixture in a quartz capillary under applied electric field.

An aliquot (1.3 mL) of the analyzed solution (100 mg of SMCs dispersed in 10 ml of H_2O) is introduced into the capillary, pre-filled with the buffer solution – electrolyte. The electrolyte contains 20 mmol/L of benzimidazole, 4 mmol/L of tartaric acid and 2.0 mmol/L of 18-crown-6. Then, high voltage (up to 30 kV) is applied to the ends of the capillary, and the components of the mixture begin to move towards the detection zone at different speeds depending on their charge and mass. Peaks on the electropherogram represent information regarding the detecting zone's passing, which is digitized and communicated to the computer. By comparison of the calibration curves with experimental measures the ammonium concentration in the aliquot can be estimated.

This method was used to detect ammonium ions diffusing into an aqueous medium from capsules with a permeable shell. Samples with loading of 100 mg microcapsules/10 ml water were used for the determination. Samples were collected at the following time intervals: 30 minutes, 1 hour, 2 hours, 3 hours.

Results and discussion

Chemical composition of microcapsules analyzed by FTIR

The FTIR spectra of APS, ABS, TMPTA crosslinked by BAPO, as well as SMCs and DMCs were obtained (Figures 3 and 4). For pure APS very strong peaks were observed on the FTIR spectrum at 1420, 1300 and 1236 cm^{-1} (Figure 4). These peaks can be attributed to O=S=O vibrations. The peaks at 1047 cm^{-1} can be attributed to the stretching vibration of S=O (Figure 4). All the peaks on the FTIR spectrum of core material are consistent with the characteristic peaks of pure APS reported in the literature [14]. Moreover, all the peaks observed on the FTIR spectrum of permeable shell material are consistent with the characteristic peaks of pure ABS (Figure 3 and 4)[46]. Registration of the FTIR spectra were collected from the surface of SMCs avoiding the crushing of them. All characteristic peaks of ABS are present on the FTIR spectrum of SMCs. However, no peaks of APS appeared on the FTIR spectrum of SMCs which indicates that the surface of the microcapsules evenly covered the APS core (Figure 3 and 4).

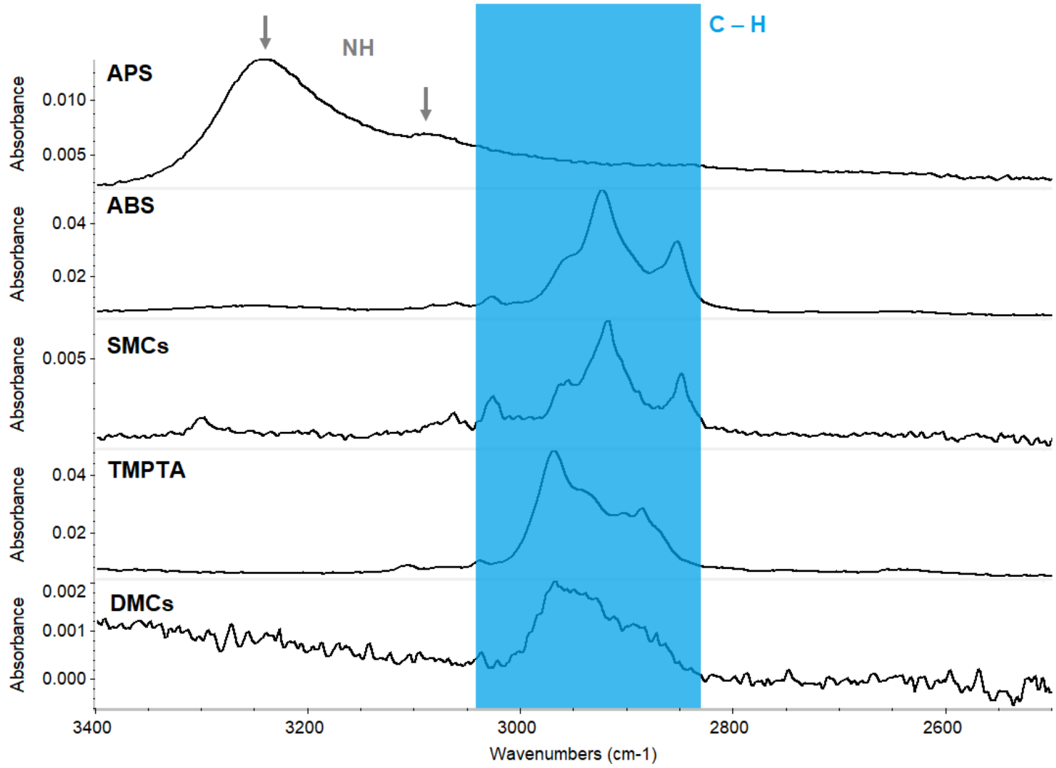


Figure 3—ATR-FTIR spectra of APS, ABS, TMPTA crosslinked by BAPO, SMCs and DMCs in a range of wavelength from 3400 to 2500 cm^{-1}. The peaks attributed to the NH group marked by grey arrows. The C-H region highlighted by blue color.

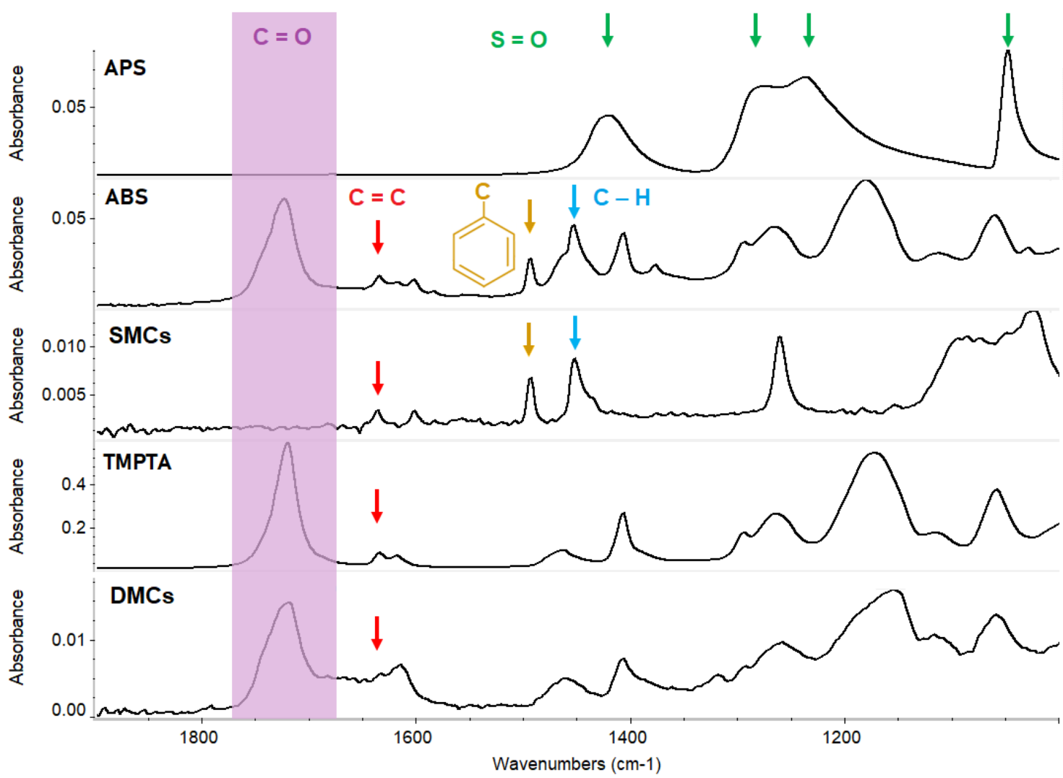


Figure 4—ATR-FTIR spectra of APS, ABS, TMPTA crosslinked by BAPO, SMCs and DMCs in a range of wavelength from 1900 to 1000 cm^{-1}. The peaks attributed to the S=O, C=C, C-H and aromatic ring groups marked by green, red, blue and orange arrows, respectively. The C=O region highlighted by violet color.

The spectrum of crosslinked TMPTA and DMCs are very similar (Figure 3 and 4). The intensity of the characteristic's peaks of -C=C- group at 1634 cm^{-1} lower for microcapsules in comparison with pure monomer supporting the crosslinking reaction of TMPTA (Figure 4). The peaks in region 3050-2800 cm^{-1} represent aromatic C-H vibrations, while the bands at 1430 cm^{-1} represent aliphatic C-H vibrations (Figure 3). The band at 1494 cm^{-1} is define as bonds in aromatic ring in styrene (Figure 4).

The results of FTIR spectrum (Figure 3 and 4) indicate that APS incorporated into first ABS shell and second TMPTA shell. The small changes on spectrum might confirm the presence of hydrogen bonding between APS and ABS and presence of water, thus elucidating the mechanism of interactions of core/ material interface [47].

Structure of microcapsules

The morphology of SMCs and DMCs microcapsules is shown in Figure 5. The SEM images (Figure 5a and 5c) shows that both microcapsules are spherical. The average particle sizes of obtained SMCs and DMCs are 80 µm and 550 µm, respectively. The surface of SMCs is porous (Figure 5(a,b)) and the pore sizes vary from 100 nm to 500 nm. The porous structure might be caused by desolvation of DCM during the solidification of coacervate ABS. The SEM image confirm the presence of a porous structure on the surface of SMCs. The SEM images of DMCs (Figure 5(c,d)) confirm the smooth and impermeable surface of them.

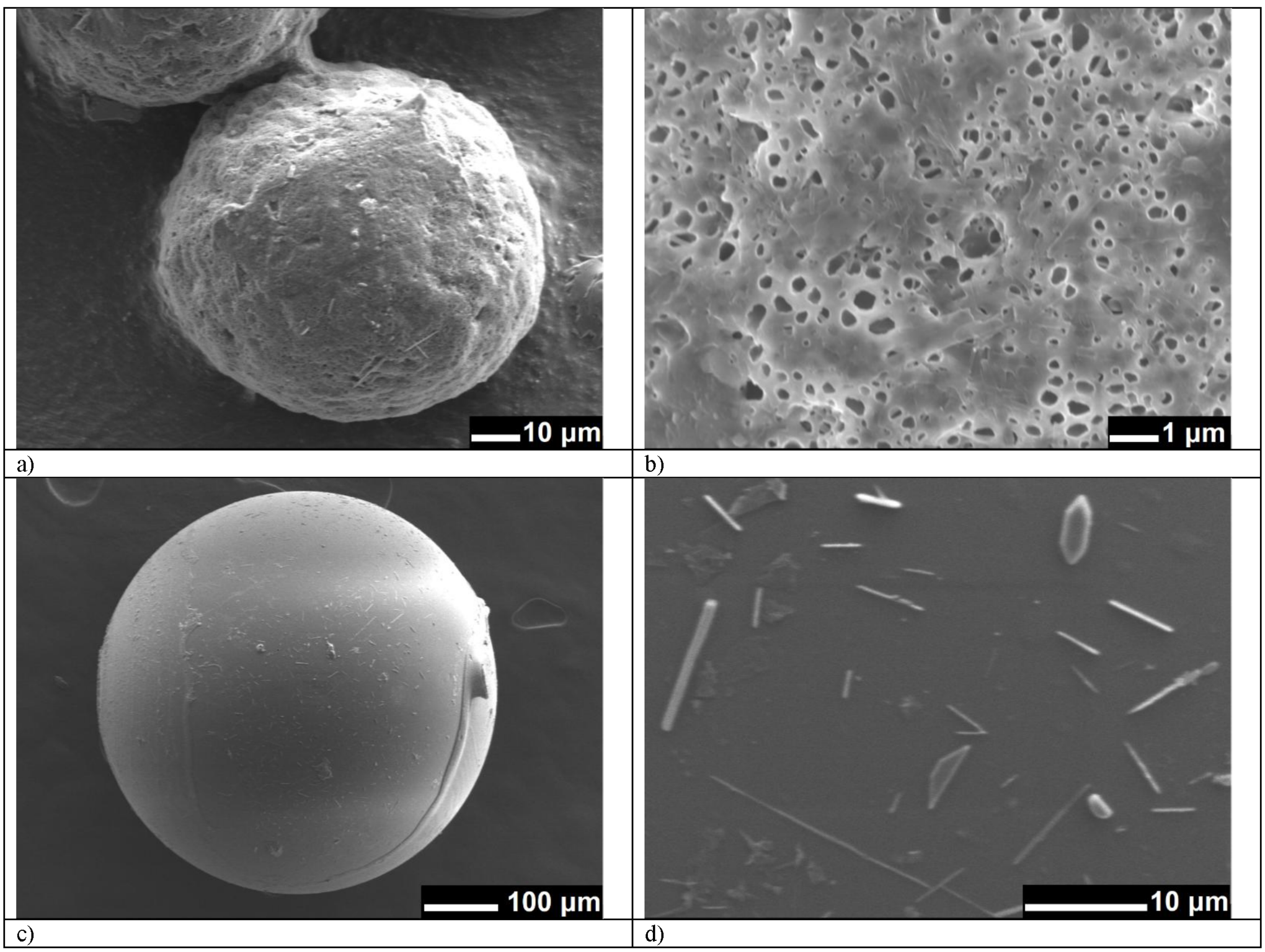
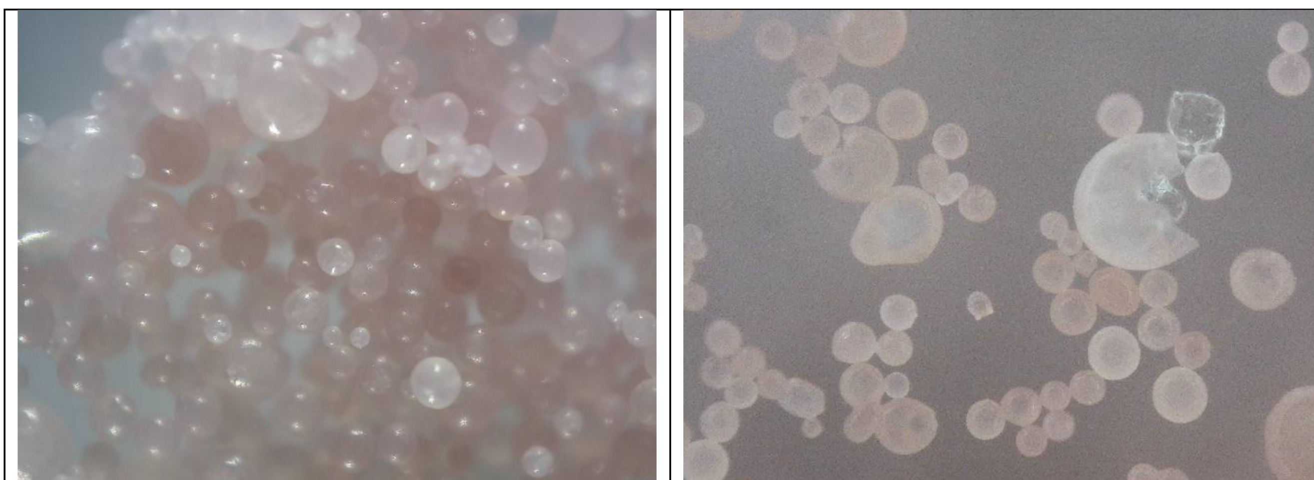


Figure 5—SEM images of SMCs (a, b) and DMCs (c, d).

To determine the size distribution and the presence of APS core inside SMCs the optical microscopy was used. The initial SMCs and SMCs after mechanical force applied are shown on Figures 6–8. The shape of the microcapsules obtained with heptane as a precipitating solvent is round with even size distribution of the capsules. However, the use of decane and low content of the PDMS leads to uneven shape and the partial aggregation of the particles. Mechanical stress would lead to the destruction of microcapsules shell and the release of the inner core. This can be observed on Fig. 6b, 7b and 8b. Transparent parts of the crushed SMCs can be attributed to the APS. It proves the fact that each microcapsules exhibit the ABS shell covering the APS particles. The average size of the particles is 80 μm.

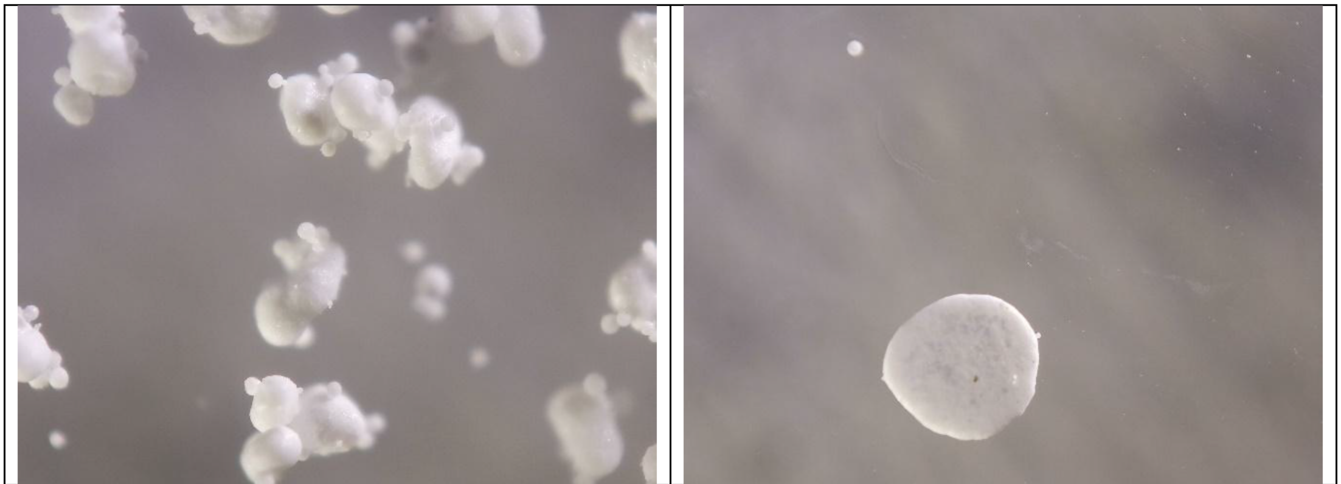
a) b)

Figure 6—Experiment N1. (a) Initial SMCs and (b) the SMCs after applied stress.



a) b)

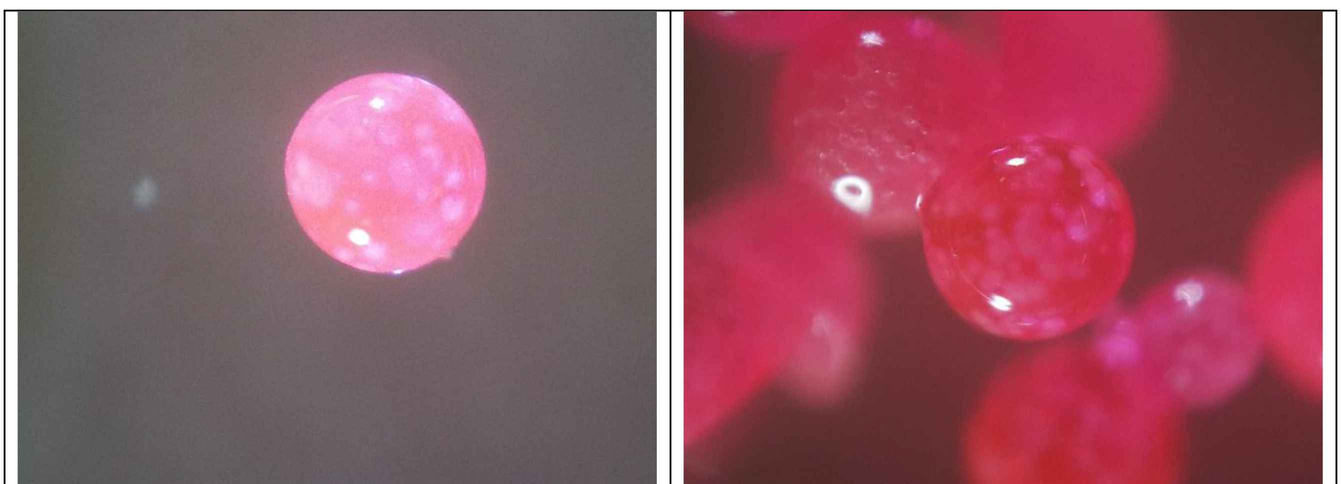
Figure 7—Experiment N2. (a) Initial SMCs and (b) the SMCs after applied stress.



a) b)

Figure 8—Experiment N3. (a) Initial SMCs and (b) the SMCs after applied stress.

The SMCs with permeable shell based on ABS were used to prepare the DMCs with outer impermeable shell by microfluidics device. As the key material for second impermeable shell the vinyl acrylate was chosen. The reason of that it has high glassy transition temperature over 400 F will result in shell with high mechanical strength and high HT stability. The Figure 9 presents the images of the DMCs. The visible white inclusions are attributed to the SMCs described above. The pink glassy shell was formed by photo-polymerization of vinyl acrylates. Pink color comes from the TMPTA phase due to addition of Rhodamine B. The average size of the DMCs is 500 µm which is in a good agreement with SEM.



a) b)

Figure 9—DMCs images of the optical microscopy.

Preparation and release mechanism of microcapsules

Figure 1 illustrates the microencapsulation procedure where APS is encapsulated via coacervation of ABS triggered by PDMS. Initially, fine APS particles are uniformly suspended in an ABS solution, where the polymer chains remain dispersed in random orientations. Upon introducing PDMS, the solubility of ABS decreases significantly, inducing phase separation and coacervation. The resulting ABS coacervate droplets exhibit adhesive properties and gradually deposit onto the APS particle surfaces, driven by interfacial tension forces and intermolecular hydrogen bonding (O-H···N interactions).

Subsequent immersion of this mixture in n-heptane rapidly solidifies the ABS coacervate through solvent diffusion, yielding fully formed SMCs microcapsules. Scanning electron microscopy (SEM) analysis reveals that the microcapsule shell, composed of polymer shell with a porous morphology (Fig. 3 a and b). When these microcapsules are introduced into an aqueous medium, water penetrates through the surface pores, dissolving the encapsulated APS core. The resulting APS solution then diffuses outward through the permeable shell matrix into the surrounding media. This diffusion-controlled mechanism enables the sustained release of APS over time.

Properties of SMCs

In the Table 1 used conditions for SMCs' preparation are presented. The following parameters were varied: APS / ABS content, precipitating solvent, PDMS amount and the water content in DCM. Additionally, the yield of the resulted SMCs is presented on the Table 1. When the anhydrous DCM used the yield of SMCs' formation can be reached to the 48 % using heptane as precipitating solvent. The conversion of the ABS into the microcapsules shell is higher when the decane is used as precipitating solvent.

Table 1—Systems for permeable shell production

Number of experiments	APS, g	ABS, g	PDMS, ml	DCM, ml	Precipitating solvent	Additional comments	Yield
N1	0.8	1	30	30	Heptane	-	33%
N2	0.8	1	30	30	Heptane	Anhydrous DCM	48%
N3	0.8	0.8	15	30	Decane	Anhydrous DCM	93%

Oxidizer release measurements

The figure 10 shows the amount of kinetics of APS release from the 100 mg of SMCs dispersed in 10 ml of H_2O with time. The samples obtained during the experiment N3 exhibit the fast release during the first 30 min, while the samples obtained during the experiments N1 / N2 are exhibit the gradual release of the oxidizer for about 6 mg/mg per 1 hour. The fast release of the samples in experiment N3 can be attributed to the malformed shell of the microcapsules. The complete release of oxidizer through permeable shell presented in the Table 2. Kinetics of complete release of APS through of permeable shell was evaluated with capillary electrophoresis. The comparison of ammonium concentration in experimental aliquots and standard calibration curves allows to determine APS content and time of complete release. The experimental results presented in Table 2. The biggest release of APS was observed for sample N3 and was 64 % of total amount of APS used for synthesis of SMCs.

Table 2—The APS releasing through permeable shell in time intervals.

System	APS releasing, mg				Theoretical loading of APS in 100 mg of capsules, mg	Total released weight of APS, mg
	After 30 min	After 1 h	After 2h	After 3h		
N1	5.74	5.12	7.74	7.35	44	25.96 (60%)
N2	13.43	2.68	4.09	1.28	44	21.49(49%)
N3	23.48	5.59	1.74	0.97	50	31.8 (64%)

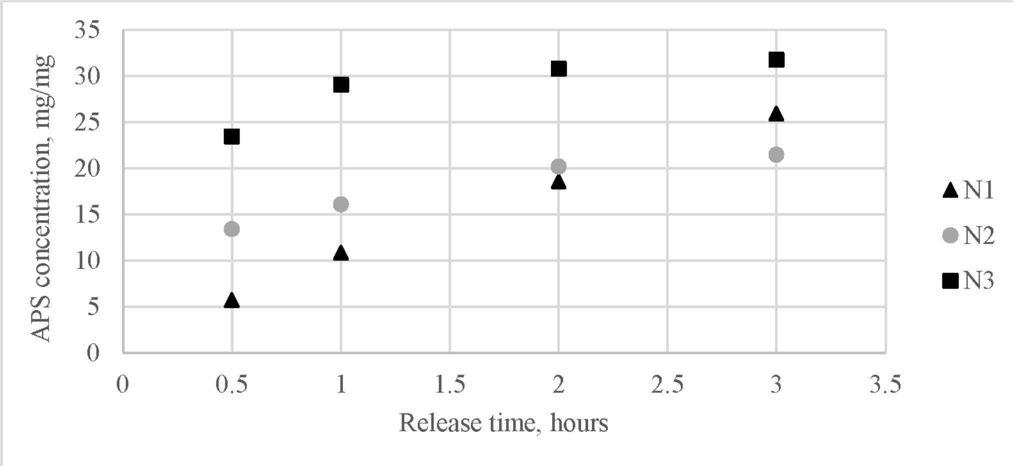


Figure 10—Total amount of the APS released from SMCs.

Conclusion

We developed the efficient technique for APS encapsulation via coacervation of ABS induced by PDMS The technique is simple and require mild synthesis conditions. The obtained SMCs exhibited a porous surface and core/shell structure. The sustained release of oxidizer can be tuned via variation of synthesis conditions. We can efficiently control the size distribution of DMCs via microfluidics technology. DMCs coated by photo-crosslinked acrylate shell exhibit high glass transition temperature which is beneficial for high-temperature fracturing operations.

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